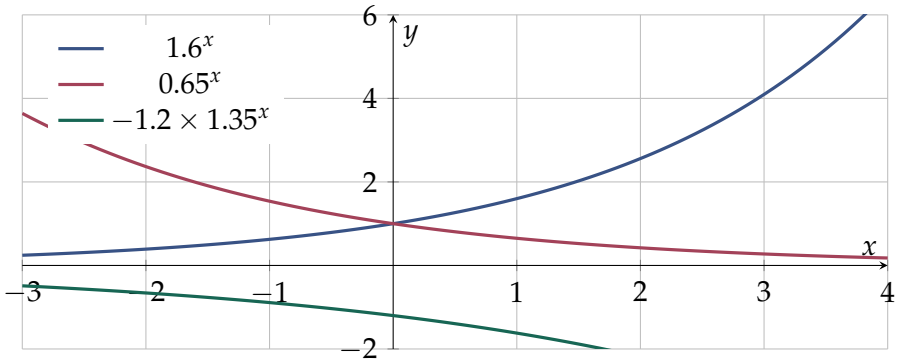
CHAPTER 2

# Exponential Functions

**Skills:** model a continuous change, switch between representations, calculate, interpret

## 2.1 Core Course Notes

For $a > 0$, the function $x \mapsto a^x$ extends the integer powers of $a$ to all real numbers. It satisfies $a^0 = 1$ and $a^{-x} = \dfrac{1}{a^x}$.




**Key point.** If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. For $f(x) = ka^x$, multiplying by $k < 0$ reverses the direction of variation. In every case, $a^x > 0$.

For $a > 0$ and all real numbers $x, y$:

$$a^{x+y} = a^x a^y, \qquad a^{x-y} = \frac{a^x}{a^y}, \qquad (a^x)^y = a^{xy}.$$

**Watch out.** In general, $a^x + a^y \neq a^{x+y}$. The rules apply to products and quotients, not to sums.

### 2.1.1 Changes and average rate

A quantity with initial value $V_0$ undergoing a constant relative change with multiplier $q$ is modelled by $V(t) = V_0 q^t$. If an overall multiplier $Q$ results from $n$ identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $t = q - 1$.

**Example 1.** A company's turnover goes from 80 000 to 95 000 euros in four years. The overall multiplier is $Q = \dfrac{95\,000}{80\,000} = 1.1875$. The average annual multiplier is $q = 1.1875^{\frac{1}{4}} \approx 1.0440$: the average annual rate is therefore about 4.40 %.

### 2.1.2 Solving and interpreting

When both sides can be written with the same base, we use injectivity: for $a > 0$ and $a \neq 1$, $a^u = a^v$ is equivalent to $u = v$. Otherwise, a graphical or numerical solution gives an approximate value; the common logarithm (base 10), studied later, will give a formula.

## Session 7 — Definition and direction of variation of $a^x$

Exponential Functions

6 Oct 2026 • B: 2 h – C: 1 h

**Objectives.**

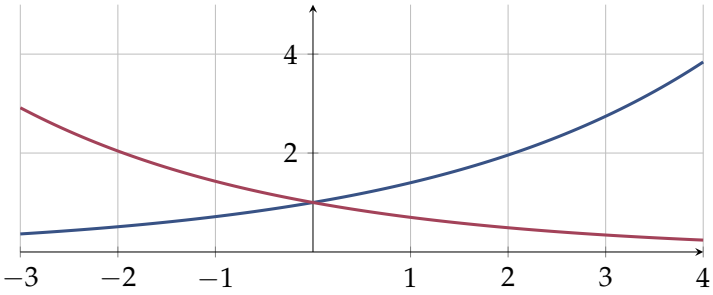
- Make sense of $a^x$.
- Link the base to the direction of variation.
- Read curves.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 10 % increase. .....
2. Compare $1.04^3$ and $1.04^7$. .....
3. Calculer $2^4 \times 2^3$. .....
4. Write $5^{-2}$ as a fraction. .....
5. Is the function $x \mapsto 0.7^x$ increasing?? .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: make sense of $a^x$ and link the base to the direction of variation.

**Activity 1.** Using a calculator, complete a table of values of $1.5^x$ and $0.7^x$ for $x = -2, -1, 0, 1, 2, 3$. Plot the points, conjecture the variations and identify the point common to both curves.




**Written record.** For $a > 0$, $a^x$ is defined for every real number $x$. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

**Example 2.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.1 · essential.**

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1.04^x$ and $g(x) = -4 \times 0.8^x$.

3. An initial value of 1350 increases by 4 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, course notes pp. 42–47; exercises p. 48, no. 1–8. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** On 1 January 2024, the capacity of the solar panels installed in a region is 240 megawatts (MW). This capacity is estimated to grow by 12 % per year. For every natural number $n$, let $u_n$ be the installed capacity, in MW, on 1 January of the year $2024 + n$; we assume that $(u_n)$ is a geometric sequence.

1. Give the common ratio of the sequence $(u_n)$ and express $u_n$ in terms of $n$.
2. We want to estimate the installed capacity at any date. Suggest an expression for a function $f$ giving the installed capacity $x$ years after 1 January 2024, where $x$ is a positive real number.
3. Estimate, to the nearest 0.1 MW, the installed capacity on 1 January 2026, on 1 April 2027 and on 1 July 2029. A quarter is taken as 0.25 year.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 155 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 8 — Powers and algebraic rules

Exponential Functions

8 Oct 2026 • B: 1 h – C: 2 h

**Objectives.**

- Transform powers.
- Use negative exponents.
- Spot a rule that does not apply.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 4 - x + 2$. .....
2. Calculer $2^1 \times 2^3$. .....
3. Write $5^{-2}$ as a fraction. .....
4. Is the function $x \mapsto 0.7^x$ increasing?? .....
5. Compare $1.04^3$ and $1.04^7$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: transform powers and use negative exponents.

**Activity 2.** Simplify $2^3 \times 2^{-5}$, $5^{x+2} \div 5^x$ and $(3^x)^2$. Then invent a numerical counterexample to $a^x + a^y = a^{x+y}$.

**Written record.** $a^{x+y} = a^x a^y$, $a^{x-y} = \dfrac{a^x}{a^y}$ and $(a^x)^y = a^{xy}$. These rules require the same positive base.

**Example 3.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.2 · essential.**

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1.05^x$ and $g(x) = -5 \times 0.8^x$.
3. An initial value of 1400 increases by 5 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, course notes pp. 43–45; exercises p. 49, no. 9–13. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** Write each of the following numbers as a single power of 3:

$$a = 3^{-2} \times 3^{6}; \qquad b = \left(3^{1.5}\right)^{4}; \qquad c = 27 \times 9^{-2}; \qquad d = \frac{3^{8}}{9^{3}}.$$

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 160 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 9 — Functions $x \mapsto ka^x$: variation and graph

Exponential Functions

13 Oct 2026 • B: 2 h – C: 1 h

**Objectives.**

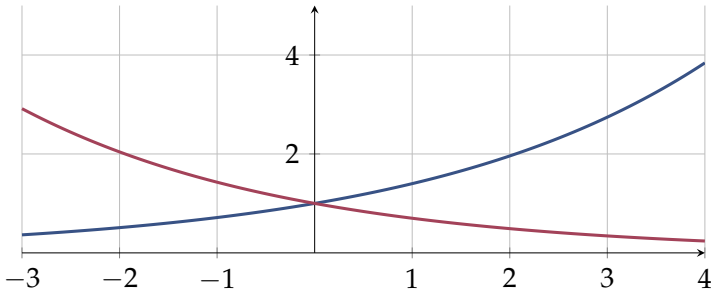
- Study $ka^x$.
- Combine sign and variation.
- Link formula and graph.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 20 % of 100. .....
2. Write $5^{-2}$ as a fraction. .....
3. Is the function $x \mapsto 0.7^x$ increasing?? .....
4. Compare $1.04^3$ and $1.04^7$. .....
5. Calculer $2^2 \times 2^3$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: study $ka^x$ and combine sign and variation.

**Activity 3.** Match the four functions $2^x$, $0.6^x$, $-2^x$ and $-0.6^x$ to four curves. Justify each choice using the sign, the $y$-intercept and the direction of variation.




**Written record.** For $f(x) = ka^x$, the sign is that of $k$. If $k < 0$, multiplying by $k$ reverses the direction of variation of $a^x$.

**Example 4.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.3 · essential.**

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1.06^x$ and $g(x) = -6 \times 0.8^x$.

3. An initial value of 1450 increases by 6 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, course notes pp. 44–47; exercises p. 48, no. 5–8, then p. 52, no. 31–36. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** Determine the direction of variation on $\mathbb{R}$ of the functions defined by:

$$f : t \mapsto 3.5 \times 0.82^t; \qquad g : t \mapsto -0.6 \times 1.15^t; \qquad h : t \mapsto -4 \times 0.7^t.$$

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 165 and the rate or common difference/ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Session 10** — **Average rate of change**

Exponential Functions — 15 Oct 2026 • B: 1 h – C: 2 h

**Objectives.**

- Calculate an average rate.
- Distinguish overall and average rates.
- Interpret an exponent $\frac{1}{n}$.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 25 % increase. .....
2. Is the function $x \mapsto 0.7^x$ increasing?? .....
3. Compare $1.04^3$ and $1.04^7$. .....
4. Calculer $2^3 \times 2^3$. .....
5. Write $5^{-2}$ as a fraction. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: calculate an average rate and distinguish overall and average rates.

**Activity 4.** The price of a service goes from 18 to 23.50 euros in five years. Calculate the overall multiplier, then the average annual multiplier and the average annual rate. Check by applying the change five times in a row.

**Written record.** If $Q$ is the overall multiplier of $n$ identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $q - 1$.

**Example 5.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.4 · essential.**

1. Simplify $3^{x+2} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 2 \times 1.02^x$ and $g(x) = -2 \times 0.8^x$.
3. An initial value of 1500 increases by 2 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, course notes pp. 47–49; exercises p. 49, no. 14–19, then p. 54, no. 47–50. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** Over five years, a factory has reduced its water consumption by 27 %.

a. Give the overall multiplier over the five years.

b. Calculate the average annual rate of change of the water consumption, to the nearest 0.1 %.

c. Check the result by applying this rounded average multiplier five times.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 170 and the rate or common difference/ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 11 — Reading graphical representations

Exponential Functions

3 Nov 2026 • B: 2 h – C: 1 h

**Objectives.**

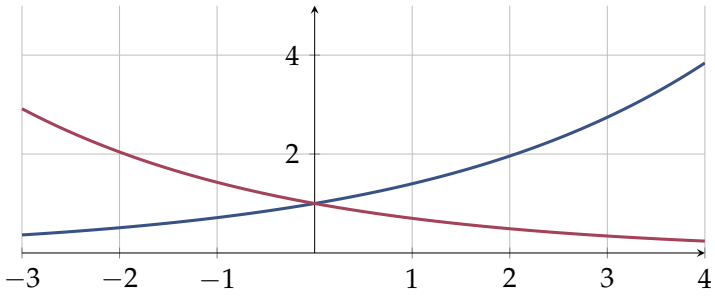
- Read outputs and find the inputs that give them.
- Compare two models.
- Estimate graphically.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 7 - x + 2$. .....
2. Compare $1.04^3$ and $1.04^7$. .....
3. Calculer $2^4 \times 2^3$. .....
4. Write $5^{-2}$ as a fraction. .....
5. Is the function $x \mapsto 0.7^x$ increasing?? .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: read outputs and find the inputs that give them and compare two models.

**Activity 5.** Two commercial offers are modelled by $f(x) = 120 \times 1.08^x$ and $g(x) = 180 \times 0.96^x$. Draw the curves on $[0, 12]$, read off their point of intersection and interpret the regions where each value is the greater.




**Written record.** Reading $f(x)$ means going from $x$ to the curve and then to the $y$-axis. Finding an input for a given $y$ means going from $y$ to the curve and then to the $x$-axis.

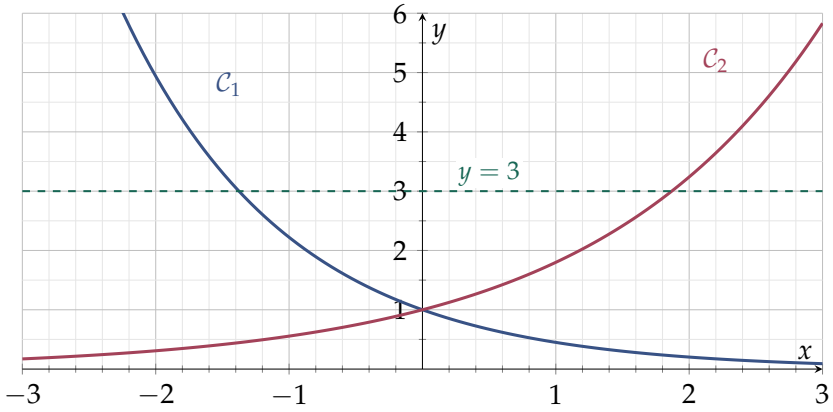
**Example 6.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.5 · essential.**

1. Simplify $3^{x+3} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 3 \times 1.03^x$ and $g(x) = -3 \times 0.8^x$.
3. An initial value of 1550 increases by 3 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, p. 52, no. 31–36. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** The graph below shows, in a coordinate system, the functions $f$ and $g$ defined on $\mathbb{R}$ by $f(x) = 1.8^x$ and $g(x) = 0.45^x$, together with the line with equation $y = 3$.




1. Which curve represents the function $f$? Justify your choice.
2. Solve the equation $f(x) = 3$ graphically. Using a table of values on the calculator, give an approximate value of the solution to the nearest 0.01.
3. Determine graphically the set of real numbers $x$ such that $f(x) \geqslant 3$, then the set of real numbers $x$ such that $g(x) \geqslant 3$.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 175 and the rate or common difference/ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Session 12** — **Simple exponential equations**

Exponential Functions — 5 Nov 2026 • B: 1 h – C: 2 h

**Objectives.**

- Solve using a common base.
- Approximate a solution.
- Check plausibility.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 5 % of 160. .....
2. Calculer $2^1 \times 2^3$. .....
3. Write $5^{-2}$ as a fraction. .....
4. Is the function $x \mapsto 0.7^x$ increasing?? .....
5. Compare $1.04^3$ and $1.04^7$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: solve using a common base and approximate a solution.

**Activity 6.** Solve $4^x = 64$, $9^x = \dfrac{1}{3}$ and $1.12^x = 2$: first using a common base when possible, then numerically for the last one. Bracket the last solution between two integers.

**Written record.** To solve $a^x = b$: use a common base when possible; otherwise bracket the solution or use a numerical tool. Check that $b > 0$.

**Example 7.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.6 · essential.**

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1.04^x$ and $g(x) = -4 \times 0.8^x$.
3. An initial value of 1600 increases by 4 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, p. 52, no. 37–38, then p. 54, no. 45–46. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** Solve the following inequalities on $\mathbb{R}$.

a. $0.7^x \leqslant 0.7^{-1}$

b. $2.5^{x+1} > 2.5^4$

c. $0.5^x \geqslant 0.25$

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 180 and the rate or common difference/ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 13 — Modelling with an exponential function

Exponential Functions

10 Nov 2026 • B: 2 h – C: 1 h

**Objectives.**

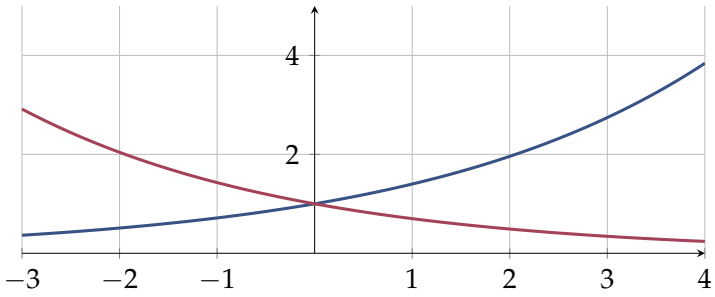
- Build an exponential model.
- Identify the parameters.
- Predict and interpret.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 10 % increase. .....
2. Write $5^{-2}$ as a fraction. .....
3. Is the function $x \mapsto 0.7^x$ increasing?? .....
4. Compare $1.04^3$ and $1.04^7$. .....
5. Calculer $2^2 \times 2^3$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: build an exponential model and identify the parameters.

**Activity 7.** An app has 12 000 users at launch and its usage grows by 9 % per month. Build $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25 000 users.




**Written record.** A constant relative change with multiplier $q$ is modelled by $f(t) = V_0 q^t$. $V_0$ is the initial value and $q$ the multiplier per unit of time.

**Example 8.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.7 · essential.**

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1.05^x$ and $g(x) = -5 \times 0.8^x$.

3. An initial value of 1650 increases by 5 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, p. 50, no. 20–25, then p. 53, no. 41. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** Just after an injection, the concentration of a drug in a patient's blood is 80 mg/L. This concentration then decreases by 15 % every hour. This is an exponential decay.

It is modelled by a function $C$ of the time $t$, in hours, such that $C(t) = ka^t$.

1. Determine the constants $k$ and $a$.
2. Calculate the concentration after 3 h 30 min, to the nearest 0.1 mg/L.
3. The doctor plans a new injection as soon as the concentration drops below 50 mg/L. Is this already the case after 3 h 30 min? Answer in a sentence.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 185 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 14 — Review and chapter assessment

Exponential Functions

12 Nov 2026 • B: 1 h – C: 2 h

**Objectives.**

- Use what the chapter has covered.
- Write up a modelling solution.
- Correct a mistake.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 3 - x + 2$. .....
2. Is the function $x \mapsto 0.7^x$ increasing?? .....
3. Compare $1.04^3$ and $1.04^7$. .....
4. Calculer $2^4 \times 2^3$. .....
5. Write $5^{-2}$ as a fraction. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: use what the chapter has covered and write up a modelling solution.

**Activity 8.** Using four pieces of student work, spot and correct: rate/multiplier confusion, a false power rule, an error in the direction of variation and a conclusion without a unit.

**Written record.** To validate an exponential model: identify $V_0$ and $q$, calculate with the units, check the order of magnitude and interpret the result.

**Example 9.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 2.8 · essential.**

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1.06^x$ and $g(x) = -6 \times 0.8^x$.
3. An initial value of 1700 increases by 6 % per year. Give the model and calculate the value after 6 years.

**Further practice:** Calao, p. 54, no. 47–58. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.**

1. The price of a digital currency rises by 125 % one year, then falls by 36 % the following year. What is the average annual rate of change over these two years?
2. The price of a raw material rises by 80 % one month, then falls by 55 % the following month. What is the average monthly rate of change?
3. A student claims that, in question 2, the average rate is $\dfrac{80-55}{2} = 12.5\,\%$. Explain in a sentence why this is wrong.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 190 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.